

GROUP10 | The Compounding Effect of Sadness & Alcohol on Youth Smoking

Research Question: Does psychological sadness have a compounding effect with substance dependency on adolescent smoking?

1 Study Framework

Group Definition

SadOrHopeless_Recoded (Binary)

Exposed Group (1): Felt sad/hopeless in past 12 months

Comparison Group (0): Did not feel sad

Response Variable:

CurrentCigaretteUse_Recoded (Binary)

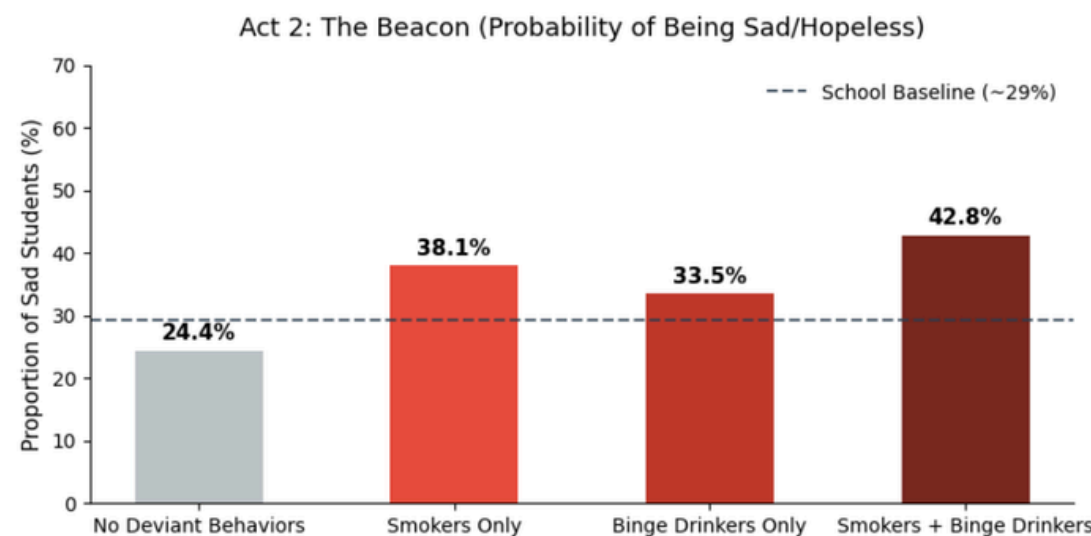
Success (1): Smoked ≥ 1 day in past 30 days

Failure (0): Smoked 0 days

2 Statistical Method

Method: Two-Proportion Z-Test

Assumption Check: Robustly passed the Success/Failure condition (All cell counts far exceed 10; Sample size $n = 13,174$).



3 Key Result

TOTAL SAMPLE (n)

13,174

Z-STATISTIC (p-value)

15.05 ($p < 0.001$)

COMPOUND RISK

4.7 x Higher

4 Conclusion

1. Shattering Rebellion Stereotypes

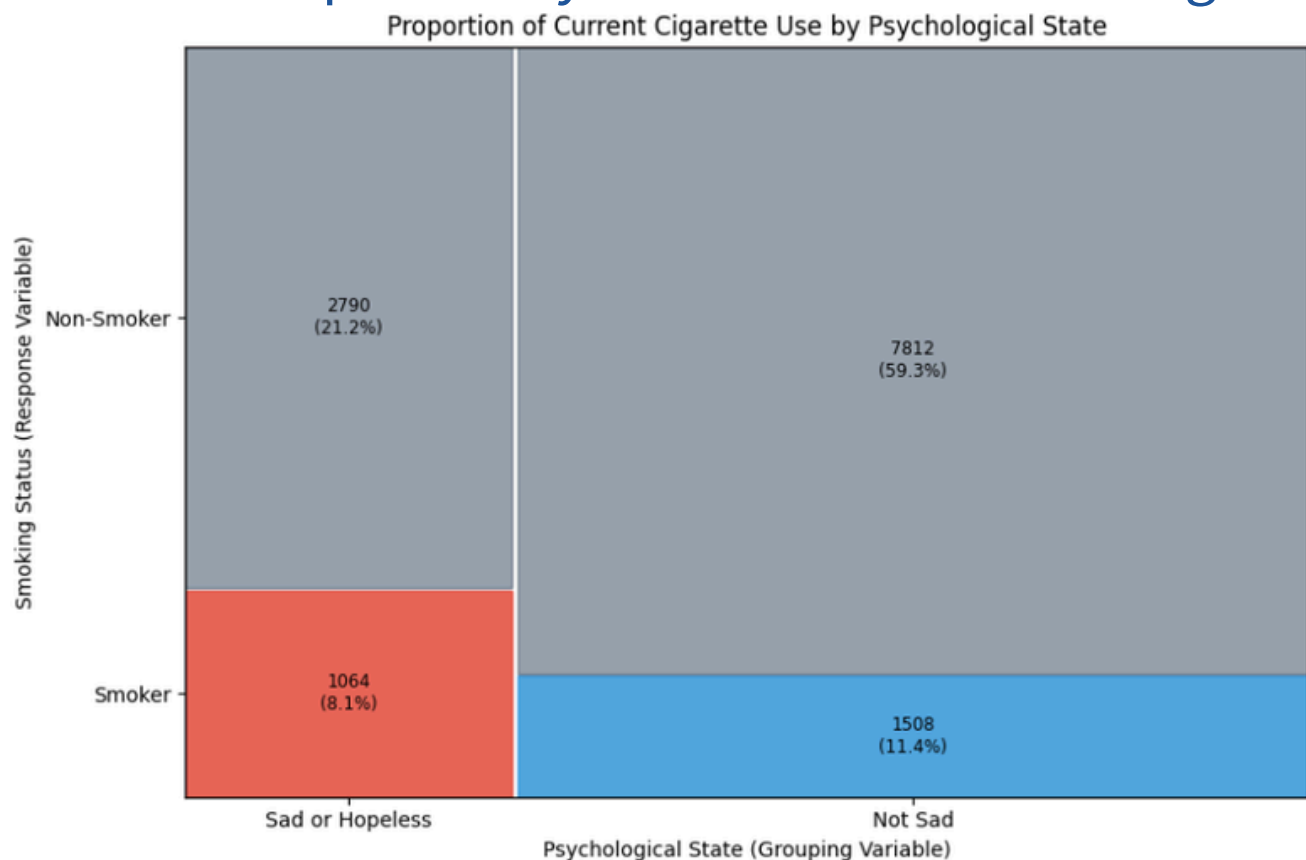
Pure rebellion accounts for only 20.1% of student smokers. The vast majority carry hidden, profound psychological or substance dependency burdens.

2. The Compounding Trap

Psychological sadness alone increases smoking rates to 27.6%. When combined with binge drinking, the rate of severe depression among smokers peaks at 42.8% (See Act 2).

3. Synergistic Risk & Integrated Policy

The overlay of sadness and alcohol dependency rockets the tobacco risk to 4.7x higher than the baseline. School campaigns must pivot from behavior isolation to integrated models combining Tobacco Cessation with Student Mental Health Support.



27.6% of students in the "Sad or Hopeless" group are smokers, compared to only 16.18% in the "Not Sad" group.